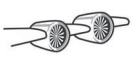
Refining crude oil

Crude oil is a **fossil fuel** that formed millions of years ago from buried dead plant and animal matter. These remains were turned into fossil fuels by **heat** and **pressure**. Crude oil is a mixture of different compounds. Most of the compounds in crude oil are **hydrocarbons**. The only atoms they contain are carbon and hydrogen.

Some common hydrocarbons found in crude oil	Chemical formula
methane	CH_4
ethane	C_2H_6
propane	C_3H_{10}
butane	C_4H_{10}

Crude oil is separated into a number of different parts, called fractions, using a process called **fractional distillation**. This involves heating the crude oil until it **evaporates** and then letting the gas **condense**. The different hydrocarbons can be separated because they have different boiling points. The fractions are then processed into more useful hydrocarbons. Crude oil provides useful substances such as petrol, waxes, plastics, detergents, bitumen and man-made fibres such as nylon.

The diagram shows the different substances formed from fractional distillation. Substances with a high boiling point condense at the bottom and substances with lower boiling points condense at the top.

	Fraction	Boiling point (°C)
	LPG (liquefied petroleum gas) (used for heating and cooking)	<25
	petrol (fuel used for cars)	25–60
	naphtha (used for making chemicals)	60–180
	paraffin (flammable waxy solid used in candles and polishes)	180–220
	diesel (liquid fuel similar to petrol, used in cars, trucks and heavy machinery)	220–250
	lubricating oil (thick oil used to reduce friction between moving parts in machinery)	300–350
	bitumen (thick black substance used on roads)	>350

Fractional distillation of crude oil

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Crude oil

Use the Resource sheet 'Refining crude oil' to answer these questions.

☞ **1** Cross out any answers that are incorrect.

Crude oil formed	from dead plants and animals millions of years ago thousands of years ago
Crude oil contains the elements	carbon oxygen hydrogen
The remains of living things are changed into fossil fuels by the effects of	air pressure heat
Most of the compounds in crude oil are	metals hydrocarbons

2 Fill in the missing letters in these important words from the Resource sheet.

c__ ud__ o__ l hy__ r __ ca __ b __ __ s d__ st __ __ __ ati __ __ b __ __ l __ __ g po__ __ts

3 List four useful substances that can be produced from crude oil.

- _____
- _____
- _____
- _____

4 Use the diagram on the Resource sheet to answer these questions.

a The different fractions in crude oil are separated based on what characteristic?

_____

b What is the boiling point of the following fractions?

lubricating oil _____ paraffin _____ petrol _____

c Which fraction has the lowest boiling point? _____

d Where does it condense in the tower? _____

e What type of fraction is used for fuel in trucks? _____

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